

Shop Assistant

Help & Reference Guide

How This App Works

- Most calculators support multiple unit systems (decimal inches, fractional inches, millimeters, feet, etc.) via a unit toggle near the top of the card.
- Many layout calculators (Picket Layout, Spindle Layout, Wall Framing, Diagonal Slats) show a live diagram, and let you select an individual piece from a dropdown to see its exact position, length, and cut angles.
- Cards with a “Saved Layouts” section let you save your current inputs under a name and reload them later.
- Cards with a “Print / Save as PDF” button generate a printable summary of that card's inputs and results, including its diagram where applicable.
- Use “Data Backup” at the top of the page (Export All Data / Import Data) to move your saved layouts and settings between devices.

01 — Saw Cuts

Setup calculations for common table saw and miter saw operations.

Board Split

Rip a board into equal-width strips, accounting for blade kerf.

Fence Setting

Find the fence position for a target finished piece width.

Dado Cut

Plan dado width/depth using multiple passes of a single blade.

Unequal Board Miter

Miter angles for joining two boards of different widths.

Crown Moulding

Spring angle and blade tilt/miter settings for crown moulding.

Polygon Cuts

Miter angle, side length, and circumradius for any regular polygon.

Diagonal Slats

Per-slat blank lengths and miter angles for diagonal slat patterns on a panel, including corner and edge pieces.

02 — Tool Adjustment

Calibration and fine-tuning calculators for shop machinery.

Tablesaw Fence

Fence offset calculations for angled/tapered cuts.

5-Cut Adjustment

Classic 5-cut method for squaring a miter/crosscut sled.

Planer Adjustment

Planer depth-of-cut and material removal tracking.

Router Shims

Shim combinations to fine-tune a router bit or fence to an exact height/offset.

Clamps & Shims

Reference calculator for clamping pressure and shim stacking.

03 — Design

Layout and proportion calculators for furniture and structural design.

Picture Framing

Mat widths and reveal for multi-mat picture frames.

Drawer Design

Drawer box dimensions accounting for slide clearance and material thickness.

Rafter Design

Rafter length, seat cut, and ridge/tail cuts for a given pitch and span.

Stairs & Steps

Rise/run, stringer layout, and code-check for stair design.

Crossbuck Design

Layout for X-brace (crossbuck) door and gate patterns.

04 — Layouts

Spacing and positioning calculators for repeating elements.

Object Spacing

Even spacing of objects across a given length.

Spindle Layout

Baluster/spindle spacing between two rails, including corner spindles.

Picket Layout

Picket spacing for rectangular fence sections and sunburst gate patterns.

Square-Up (3-4-5 Method)

Verify a corner is square using the 3-4-5 triangle method, with tape-measure running totals.

Wall Framing

Stud layout for straight and raked (gable-end) walls, including corner framing.

Wheel Segments

Segment angles and lengths for a segmented wheel/circle glue-up.

Circular Pattern

Even angular spacing of items around a circle.

Rectangular Pattern

Grid spacing of items across a rectangular area.

Interval Layout

General repeating-interval spacing calculator.

05 — Quantities

Material takeoff calculators for estimating job quantities.

Area Shape

Area and perimeter for common shapes.

Wall Shape

Wall area accounting for door/window openings.

Studs

Stud count for a given wall length and spacing.

Joists

Joist count and span for floor/ceiling framing.

Decking

Decking board quantity and layout for a deck surface.

Roofing

Roofing material quantities based on roof area and pitch.

Suspended Ceiling

Grid and tile quantities for a drop ceiling.

Brick & Block

Brick/block count for a wall of given dimensions.

Drape Panels

Fabric panel quantities for window treatments.

06 — Calculators

General-purpose reference calculators used across many projects.

Unit Convert

Live conversion between decimal inches, fractional inches, millimeters, feet, meters, gallons/quarts/pints, and pints/ounces/drops.

Object Clearance

Required clearance between objects.

Shelf Sag

Predicted shelf deflection under load for a given material and span.

Cabinet Stability

Tip-over risk estimate for a cabinet based on weight and dimensions.

Golden Ratio

Golden-ratio proportion calculator for aesthetic layout.

Date Math

Date arithmetic (adding/subtracting days, finding intervals).

Complementary Angle

Complementary and supplementary angle lookup.

Slopes

General slope/angle reference calculator.

Wrench Selection

Wrench size lookup for common fastener sizes.

Slope / Grade

Convert between slope ratio, degrees, and percent grade.

Stair Rise & Run

Quick rise/run calculation for stairs.

Feet-Inches ■ Decimal

Convert between feet-inches and decimal feet/inches.

Spirit Levels

Bubble-level accuracy/reading reference.

Survey Levels

Elevation calculations for a surveying level, in decimal feet, feet-inch, or millimeters.

Wire Sizing

Wire gauge selection based on amperage, distance, and voltage drop.

Curve Solver

General curve/arc geometry solver.

07 — Data & Information

Reference tables for common hardware and building specifications.

Wood Screws

Wood screw size and pilot-hole reference.

Tap Drill

Tap drill size reference for cutting internal threads.

Kreg Screws

Kreg pocket-hole screw selection by material thickness.

Lag Bolts

Lag bolt sizing and pilot-hole reference.

Fasteners

General fastener reference table.

Drill Bits

Drill bit size reference and lookup.

Furniture Dimensions

Standard/ergonomic furniture dimension reference.

Span Tables

Structural span table reference for joists/rafters.

08 — Paint & Chemical Mixes

Mixing and dilution calculators for finishes and chemicals.

Shellac

Shellac cut ratios and dilution, including a Flakes/Dilute mode.

Latex

Latex paint mixing/dilution reference.

Wood Stains

Saved database of wood stain colors and brands.

Stain Modification

Adjusting an existing stain mix toward a target result.

Chemical Mixes

Saved database of chemical dilution ratios (e.g. cleaners).

Color Mixing

Suggests up to 3 base colors (from Red, Orange, Yellow, Green, Blue, Indigo, Violet, Purple, White, Black) and mixing percentages/ounces to approximate a target hex color. This is a simplified linear-blend approximation, not a true pigment simulation.

09 — Settings & Help

App-wide settings and this help reference.

Colors

Change the app's color theme. Saved automatically and applies everywhere.

Help

This document.